Used Car Price Prediction –

Detailed Problem Statement

1. Introduction

The automobile industry has experienced significant growth in the used car market due to increasing demand for affordable vehicles. Determining the correct price of a used car is a challenging task because the price depends on various factors such as car brand, model, manufacturing year, kilometers driven, fuel type, transmission type, ownership history, and vehicle condition.

Traditional methods of used car price estimation rely on manual inspection, dealer experience, and market comparison. These methods can be time-consuming and may not always provide accurate results.

The Used Car Price Prediction System uses Artificial Intelligence (AI) and Machine Learning (ML) techniques to predict the approximate selling price of a used car based on historical car data. The system helps users, buyers, and sellers make better decisions by providing a data-driven price estimation.

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2. Problem Definition

Predicting the accurate price of a used car is a complex problem because multiple factors influence its market value. Customers often face difficulty in identifying whether a car is fairly priced or overpriced.

The main objective of this project is to develop a machine learning-based system that can automatically predict the resale price of a used car using important features such as:

- Car brand and model
- Manufacturing year
- Present car price
- Number of kilometers driven
- Fuel type (Petrol/Diesel/CNG)
- Seller type (Dealer/Individual)
- Transmission type (Manual/Automatic)
- **Number of previous owners**
- **Vehicle age**

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3. Existing System

In the existing system, used car prices are estimated using traditional approaches:

- ### Manual Price Estimation:

- Dealers and sellers manually calculate car prices based on experience.
- Price estimation depends on personal judgment.

- ### Online Market Comparison:

- Users compare prices from different websites and advertisements.
- It requires time and effort to analyze multiple listings.

- ### Limitations of Existing System:

- Less accuracy due to human assumptions.

- Time-consuming process.
- Difficult to consider all factors affecting car prices.
- Price variations between different sellers.
- Lack of automated prediction capability.

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4. Proposed System

The proposed system introduces an AI-based Used Car Price Prediction System that uses Machine Learning algorithms to estimate car prices automatically.

The system collects historical used car data and trains a machine learning model to identify relationships between car features and prices.

➢ Features of Proposed System:

- Accepts car details from users through a web interface.
- Processes input data using machine learning techniques.
- Predicts the estimated resale value of the car.
- Provides quick and accurate results.
- Reduces dependency on manual evaluation.

➢ Technologies Used:

- Programming Language: Python
- Machine Learning Algorithm: Multiple Linear Regression / Random Forest Regression
- Libraries: Pandas, NumPy, Scikit-learn, Matplotlib
- Web Framework: Flask
- Frontend: HTML, CSS, JavaScript

- Dataset: CarDekho Used Car Dataset

Advantages:

- Faster price prediction.
- Improved accuracy compared to manual methods.
- Easy to use.
- Helps buyers and sellers make informed decisions.
- Reduces pricing errors in the used car market.

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5. Conclusion

The Used Car Price Prediction System provides an efficient and intelligent solution for estimating the resale price of used vehicles. By using machine learning algorithms, the system analyzes different car features and predicts an accurate approximate price based on historical data.

This project reduces the limitations of traditional manual price estimation methods by providing faster, reliable, and data-driven results. It helps buyers and sellers understand the fair market value of a vehicle and supports better decision-making.

The system can be further improved by using larger datasets, advanced machine learning algorithms, and real-time market data to increase prediction accuracy. Overall, the proposed system demonstrates the effective use of Artificial Intelligence and Machine Learning in the automobile industry for automated price prediction.